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CHAPTER 1 • NUMBER • TOPIC 5

Fractions

Mixed numbers into improper fractions first — every time. Then one rule per operation, and “show that” ends on exactly the given answer.



$$\frac{3}{10} = 0.3 =$$

30%**6**

VARIANTS

17

PAST-PAPER QUESTIONS

43

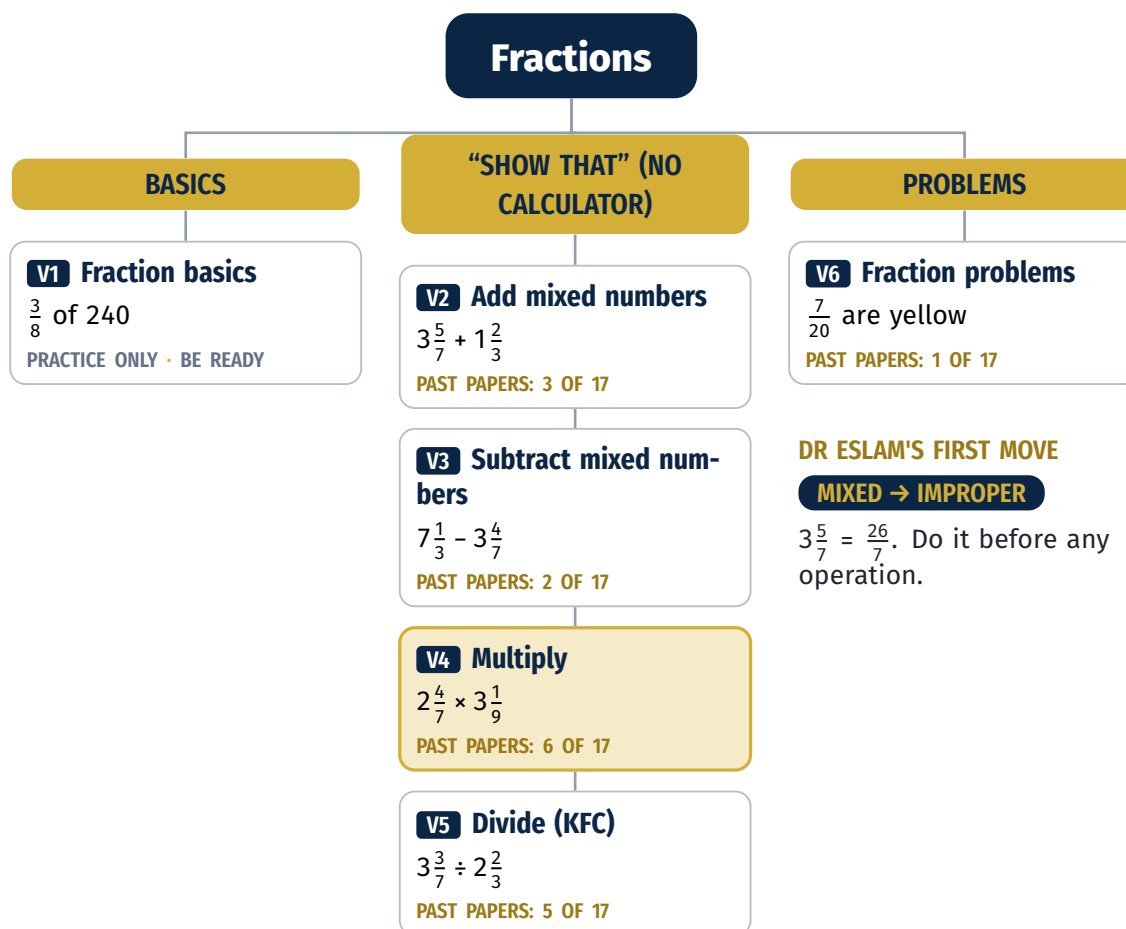
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Fraction basics **V2** Add mixed numbers **V3** Subtract mixed numbers**V4** Multiply **V5** Divide (KFC) **V6** Fraction problems

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Fractions at a glance



Your notes


How often does each variant come up?

17 past-paper questions (Jan 2020 → Nov 2025); 43 practice parts — grouped by type.

	PAST PAPERS (OUT OF 17)		PRACTICE PARTS
V1 Fraction basics	0		7
V2 Add mixed numbers	3		8
V3 Subtract mixed numbers	2		6
V4 Multiply	6		8
V5 Divide (KFC)	5		9
V6 Fraction problems	1		5

WHAT THIS TELLS YOU 16 of 17 past-paper questions are “Show that $a\frac{b}{c} \square d\frac{e}{f} = \dots$ ” on the **non-calculator** paper — 3 marks, Question 1–6. Same method every time.

BEFORE YOU START

The big idea



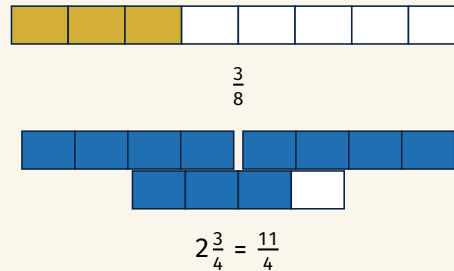
Your notes

A fraction is equal parts of a whole.

$\frac{3}{8}$: the whole split into 8 equal parts, take 3.

A **mixed number** $2\frac{3}{4}$ is wholes plus a fraction. As an **improper fraction**:

$$\frac{2 \times 4 + 3}{4} = \frac{11}{4}$$


 One rule per operation

+

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$$

same bottom first

–

$$\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$$

same bottom first

×

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

top × top, bottom
× bottom

÷ **KFC**

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Keep · Flip ·
Change

 “Show that” – the three stages the examiner wants

1 IMPROPER

$$3\frac{5}{7} = \frac{26}{7}$$

2 OPERATE

common bottom / KFC

3 BACK TO MIXED

= the given answer



CALCULATOR FIRST, THEN CHECK

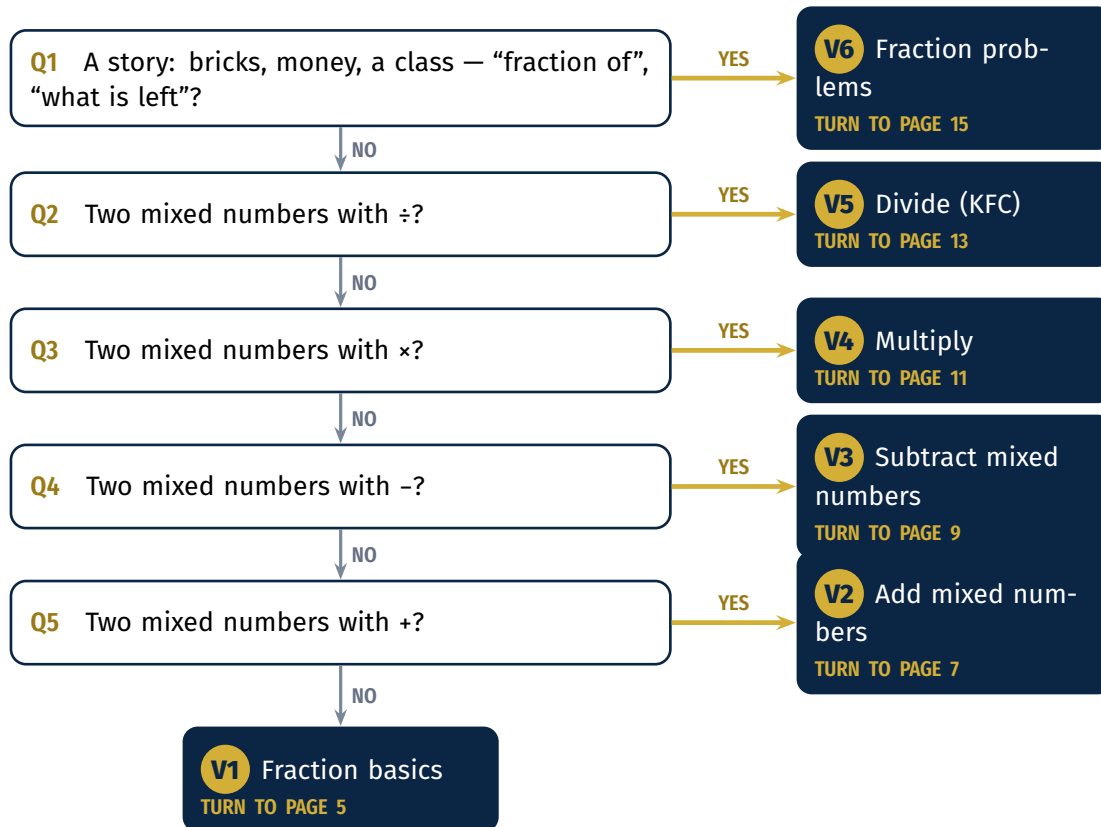
Paper 1 has no calculator — but at home use $\boxed{a^b/c}$ to check: type the question, press $\boxed{=}$, then $\boxed{\text{SHIFT}} \boxed{\text{S} \leftrightarrow \text{D}}$. In the exam, check by estimating: $3\frac{5}{7} + 1\frac{2}{3} \approx 3.7 + 1.7 = 5.4$, and $5\frac{8}{21} \approx 5.4 \checkmark$.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Fraction Basics

Mixed $\leftrightarrow$ improper, simplify, and a fraction of an amount.

SEEN IN PAST PAPERS

0 / 17

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 7


Your notes

? What is it?

In plain words

Fraction of an amount: $\frac{3}{8}$ of 240 —

divide by the bottom,

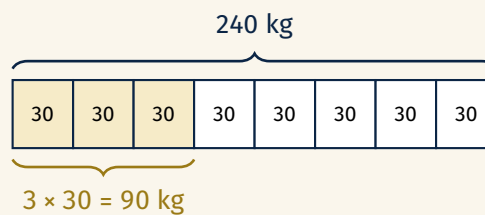
multiply by the top.

Simplify: divide top and bottom by the same number.

Mixed $\rightarrow$ improper: wholes $\times$ bottom + top.

IT LOOKS LIKE

 $\frac{3}{8}$ of 240

 $\frac{18}{24}$
 $3\frac{7}{8}$


How do I do it?

1 $\div$ the bottom

$240 \div 8 = 30$ — one part.

2 $\times$ the top

$30 \times 3 = 90$ — three parts.

3 Simplify

Divide top and bottom by their HCF.

4 Mixed $\leftrightarrow$ improper

$3\frac{7}{8} = \frac{3 \times 8 + 7}{8} = \frac{31}{8}$.

$\div$ BOTTOM, $\times$ TOP "Of" means multiply.

V1 Fraction Basics

VARIANT 1 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

“Work out $\frac{3}{8}$ of 240 kg” → $240 \div 8 \times 3$

“Write in its simplest form” → **Divide top and bottom by the HCF**

“Write $3\frac{7}{8}$ as an improper fraction” → $\frac{3 \times 8 + 7}{8}$

WORKED EXAMPLE DR ESAM

Fraction of an amount • 2 marks

Work out $\frac{3}{8}$ of 240 kg.

STEP 1 $240 \div 8 = 30$ ◀ one part

STEP 2 $30 \times 3 = 90$ ◀ three parts

ANSWER 90 kg

✓ 90 is less than 240 ✓

WATCH OUT✗ $240 \div 3 \times 8 = 640$ ✓ $240 \div 8 \times 3 = 90$

Divide by the **bottom**. A fraction less than 1 of an amount is smaller than the amount.

EXAM TIP

Simplify fully: $\frac{18}{24} \rightarrow \frac{3}{4}$ (divide by 6, the HCF, in one go).

Check yourself

calculator first • answers on page 17

1.1 Dr EsamWrite $3\frac{7}{8}$ as an improper fraction.

Answer:

1.2 Dr EsamWork out $\frac{2}{5}$ of 35 kg.

Answer:

1.3 Dr EsamSimplify $\frac{18}{24}$ fully.

Answer:

V2

VARIANT 2

Add Mixed Numbers

Improper, common bottom, add the tops,
back to a mixed number.

SEEN IN PAST PAPERS

3 / 17

PRACTICE QUESTIONS: 8


Your notes

Came up in: Jan 2020 P2H Q3 • Jun 2022 P1HR Q6 • Nov 2025 P1H Q5

? What is it?**In plain words**

You can only add fractions with the **same bottom**. Change both to the **LCM** of the bottoms (often bottom $\times$ bottom), then add the tops.

Change mixed numbers to improper first — then there is nothing to forget.

IT LOOKS LIKE

$$3\frac{5}{7} + 1\frac{2}{3}$$

$$4\frac{2}{3} + 3\frac{4}{5}$$

$$\frac{26}{7} + \frac{5}{3}$$

↓ **BOTTOM** $7 \times 3 = 21$

$$\frac{78}{21} + \frac{35}{21} = \frac{113}{21}$$

$$5\frac{8}{21}$$

How do I do it?**1 Improper**

$$3\frac{5}{7} = \frac{26}{7}, 1\frac{2}{3} = \frac{5}{3}$$

2 Common bottom

LCM of 7 and 3 is 21: $\frac{78}{21}, \frac{35}{21}$

3 Add the tops

$\frac{113}{21}$ — the bottom stays.

4 Back to mixed

$113 \div 21 = 5 \text{ r } 8$: $5\frac{8}{21}$ — matches the question.

SAME BOTTOM, ADD TOPS

Never add the bottoms.

V2 Add Mixed Numbers

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$ "

→ Improper → common bottom → mixed

"You must write down all the stages"

→ Show both converted fractions

The answer is given

→ Your last line must be exactly that

**WORKED EXAMPLE**

PAST PAPER

November 2025 · Paper 1H · Q5 · 3 marks

Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

STEP 1 $\frac{26}{7} + \frac{5}{3}$ ◀ improper

STEP 2 $\frac{78}{21} + \frac{35}{21}$ ◀ common bottom 21

STEP 3 $\frac{113}{21}$ ◀ add the tops

STEP 4 $= 5\frac{8}{21}$ ◀ $113 = 5 \times 21 + 8$

ANSWER $5\frac{8}{21}$ (as required)

✓ estimate $3.7 + 1.7 \approx 5.4$ ✓

**WATCH OUT**

✗ $\frac{5}{7} + \frac{2}{3} = \frac{7}{10}$
 ✓ $\frac{15}{21} + \frac{14}{21} = \frac{29}{21}$

Adding tops and bottoms is wrong — same bottom first.

**EXAM TIP**

"Show that": the answer is given, so the marks are for the stages. Write the improper fractions and the common-bottom line.

**Check yourself**

calculator first · answers on page 17

2.1

Jan 2020 P2H

Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$.

Answer:

2.2

Jun 2022 P1HR

Show that $2\frac{2}{3} + 3\frac{3}{4} = 6\frac{5}{12}$.

Answer:

2.3

Dr Eslam

Show that $2\frac{1}{3} + 1\frac{3}{4} = 4\frac{1}{12}$.

Answer:

V3

VARIANT 3

Subtract Mixed Numbers

Same as adding — and improper fractions avoid “borrowing”.

SEEN IN PAST PAPERS

2 / 17

PRACTICE QUESTIONS: 6



Your notes

Came up in: Jun 2022 P2H Q4 · Jun 2025 P2H Q3

? What is it?

In plain words

$7\frac{1}{3} - 3\frac{4}{7}$: the second fraction is bigger ($\frac{4}{7} > \frac{1}{3}$), which makes whole-number methods messy.

Improper fractions remove the problem: $\frac{22}{3} - \frac{25}{7}$, common bottom, subtract the tops.

IT LOOKS LIKE

$$7\frac{1}{3} - 3\frac{4}{7}$$

$$5\frac{1}{3} - 2\frac{6}{7}$$

$$\frac{22}{3} - \frac{25}{7}$$

↓ BOTTOM 21

$$\frac{154}{21} - \frac{75}{21} = \frac{79}{21}$$

$$3\frac{16}{21}$$

How do I do it?

1 Improper

$$7\frac{1}{3} = \frac{22}{3}, 3\frac{4}{7} = \frac{25}{7}.$$

2 Common bottom

$$\frac{154}{21} - \frac{75}{21}.$$

3 Subtract the tops

$$\frac{79}{21}.$$

4 Back to mixed

$$79 = 3 \times 21 + 16: 3\frac{16}{21}.$$

IMPROPER FIRST — NO BORROWING Safe every time.

V3 Subtract Mixed Numbers

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO“Show that $7\frac{1}{3} - 3\frac{4}{7} = 3\frac{16}{21}$ ”

→ Improper → common bottom → mixed

The second fraction is bigger

→ Improper fractions — no borrowing needed

“Give your answer as a mixed number”

→ Divide: whole part and remainder

WORKED EXAMPLE **PAST PAPER**

June 2025 · Paper 2H · Q3 · 3 marks

Show that $7\frac{1}{3} - 3\frac{4}{7} = 3\frac{16}{21}$ **STEP 1**

$$\frac{22}{3} - \frac{25}{7}$$

◀ improper

STEP 2

$$\frac{154}{21} - \frac{75}{21}$$

◀ common bottom 21

STEP 3

$$\frac{79}{21}$$

◀ subtract the tops

STEP 4

$$= 3\frac{16}{21}$$

◀ $79 = 3 \times 21 + 16$ **ANSWER** $3\frac{16}{21}$ (as required)✓ estimate $7.3 - 3.6 \approx 3.7$ ✓**WATCH OUT** $\times 7 - 3 = 4$ and $\frac{1}{3} - \frac{4}{7}$... stuck

✓ improper fractions first

Splitting into wholes and parts fails when the second part is bigger.

**EXAM TIP**Check the improper fraction: $\frac{22}{3} = 7\frac{1}{3}$ because $7 \times 3 + 1 = 22$.**Check yourself**

calculator first · answers on page 17

3.1

Jun 2022 P2H

Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$.

Answer:

3.2

Dr Eslam

Work out $3\frac{2}{5} - 1\frac{3}{4}$ as a mixed number.

Answer:

3.3

Practice · Hard

Work out $5\frac{2}{3} - 2\frac{3}{4}$ as a mixed number.

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

Multiply Mixed Numbers

Improper, cancel, then top × top and bottom × bottom.

SEEN IN PAST PAPERS

6 / 17

PRACTICE QUESTIONS: 8



Your notes

Came up in: Jan 2020 P1HR Q2 • Oct/Nov 2020 P1HR Q1 • Jan 2021 P1HR Q1 • Jun 2024 P2H Q4 • Nov 2024 P2H Q1 • Jun 2025 P1HR Q3

? What is it?

In plain words

No common bottom needed.
Improper first, then **cancel** any top with any bottom, then multiply tops and bottoms.
Cancelling first keeps the numbers small.

IT LOOKS LIKE

$$2\frac{4}{7} \times 3\frac{1}{9}$$

$$3\frac{1}{5} \times 2\frac{5}{8}$$

$$\begin{array}{c} 2 \\ \cancel{18}^{\cancel{2}} \times \frac{28}{\cancel{9}_1^{\cancel{4}}} \\ \cancel{7}_1^{\cancel{1}} \end{array}$$

$$\frac{2 \times 4}{1 \times 1} = 8$$

18 with 9 (÷9), 28 with 7 (÷7).

How do I do it?

1 Improper

$$2\frac{4}{7} = \frac{18}{7}, \quad 3\frac{1}{9} = \frac{28}{9}$$

2 Cancel

A top and a bottom share a factor: divide both.

3 Multiply

Top × top, bottom × bottom.

4 Mixed

Back to a mixed number (or a whole number).

CANCEL BEFORE YOU MULTIPLY

Diagonal cancelling is allowed for ×.

V4 Multiply Mixed Numbers

VARIANT 4 · SEE IT DONE, THEN TRY


 Your notes

WHEN YOU SEE → YOU DO

"Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$ "

→ Improper → cancel → multiply

× between mixed numbers

→ Never multiply the wholes and fractions separately

"all the stages in your working"

→ Write the improper and cancelled lines



WORKED EXAMPLE PAST PAPER

June 2024 · Paper 2H · Q4 · 3 marks

Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

STEP 1 $\frac{18}{7} \times \frac{28}{9}$

◀ improper

STEP 2 $\frac{2}{1} \times \frac{4}{1}$

◀ cancel 18 with 9, 28 with 7

STEP 3 $= 8$

◀ multiply

ANSWER 8 (as required)

✓ $\frac{504}{63} = 8$ without cancelling



WATCH OUT

× $2 \times 3 + \frac{4}{7} \times \frac{1}{9} = 6\frac{4}{63}$
 ✓ $\frac{18}{7} \times \frac{28}{9} = 8$

Mixed numbers can't be multiplied part by part — improper first.



EXAM TIP

6 of the 17 past-paper questions. If you don't cancel, simplify at the end: $\frac{504}{63} = 8$.



Check yourself

calculator first · answers on page 17

4.1

Jan 2020 P1HR

Show that $3\frac{1}{5} \times 2\frac{5}{8} = 8\frac{2}{5}$.

Answer:

4.2

Nov 2024 P2H

Show that $1\frac{5}{7} \times 2\frac{3}{16} = 3\frac{3}{4}$.

Answer:

4.3

Oct/Nov 2020 P1HR

Show that $3\frac{3}{4} \times \frac{7}{9} = 2\frac{11}{12}$.

Answer:

V5

VARIANT 5

Divide Mixed Numbers (KFC)

Improper, then Keep · Flip · Change — it becomes a multiply.

SEEN IN PAST PAPERS

5 / 17

PRACTICE QUESTIONS: 9



Your notes

Came up in: May 2021 P2H Q2 · Jan 2022 P2H Q2 · Jun 2023 P2H Q1 · Nov 2023 P2H Q6 · May 2024 P1HR Q4

? What is it?**In plain words**

Dividing by a fraction is

multiplying by it flipped. Dr Eslam's**KFC**: Keep the first, Flip the second, Change $\div$ to $\times$.

Improper first — never flip a mixed number.



$$\frac{24}{7} \div \frac{8}{3} \longrightarrow \frac{24}{7} \times \frac{3}{8}$$

IT LOOKS LIKE

$$3\frac{3}{7} \div 2\frac{2}{3}$$

$$2\frac{1}{3} \div 5\frac{1}{4}$$

**How do I do it?****1 Improper**

$$3\frac{3}{7} = \frac{24}{7}, \quad 2\frac{2}{3} = \frac{8}{3}$$

2 KFC

$$\frac{24}{7} \times \frac{3}{8} \text{ — only the second fraction flips.}$$

3 Cancel, multiply

$$\frac{3}{7} \times \frac{3}{1} = \frac{9}{7}$$

4 Mixed

$$\frac{9}{7} = 1\frac{2}{7}$$

KFC: KEEP · FLIP · CHANGE

Flip only the divisor.

V5 Divide Mixed Numbers (KFC)

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO"Show that $3\frac{3}{7} \div 2\frac{2}{3} = 1\frac{2}{7}$ "

→ Improper → KFC → cancel → mixed

 $\div$ between fractions

→ Keep · Flip · Change

Answer smaller than 1, e.g. $\frac{4}{9}$

→ Fine — a small number divided by a bigger one

**WORKED EXAMPLE****PAST PAPER**

November 2023 · Paper 2H · Q6 · 3 marks

Show that $3\frac{3}{7} \div 2\frac{2}{3} = 1\frac{2}{7}$

STEP 1 $\frac{24}{7} \div \frac{8}{3}$ ◀ improper

STEP 2 $\frac{24}{7} \times \frac{3}{8}$ ◀ KFC

STEP 3 $\frac{3}{7} \times \frac{3}{1} = \frac{9}{7}$ ◀ cancel 24 with 8, multiply

STEP 4 $= 1\frac{2}{7}$ ◀ back to mixed

ANSWER $1\frac{2}{7}$ (as required)✓ estimate $3.4 \div 2.7 \approx 1.3$ ✓**WATCH OUT**

× flipping the first fraction

✓ flip the **second**: $\frac{24}{7} \times \frac{3}{8}$

Keep the first exactly as it is.

**EXAM TIP**

Write the KFC line on its own — it is the method mark.

**Check yourself**

calculator first · answers on page 17

5.1

Jun 2023 P2H

Show that $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$.

Answer:

5.2

May 2024 P1HR

Show that $2\frac{1}{3} \div 5\frac{1}{4} = \frac{4}{9}$.

Answer:

5.3

Jan 2022 P2H

Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$.

Answer:

V6

VARIANT 6

Fraction Problems

A story with fractions: find what fraction is known, find one part, then the rest.

SEEN IN PAST PAPERS

1 / 17

PRACTICE QUESTIONS: 5


 Your notes

Came up in: Nov 2025 P2H Q3

? What is it?

In plain words

Find the fraction that matches a **known number**: " $\frac{7}{20}$ are yellow" → red + blue are $\frac{13}{20}$, and you know there are 52 of them.

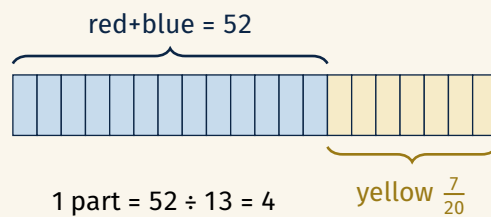
Then **one part** = $52 \div 13$, and the whole = 20 parts.

IT LOOKS LIKE

 $\frac{7}{20}$

are yellow

\$70 is left



How do I do it?

1 Known fraction

Which fraction matches a number you know? (1 – the given fraction.)

2 One part

Known number $\div$ its number of parts.

3 Scale up

One part $\times$ the parts you need.

4 Finish

The quantity asked for (weight, money ...), with units.

FIND ONE PART Then any number of parts.

V6 Fraction Problems

VARIANT 6 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“ $\frac{7}{20}$ of the bricks are yellow” + other counts

→ The others are $\frac{13}{20}$

“He has \$70 left”

→ Left fraction = 1 – spent fractions

“Work out the total weight”

→ Count × weight each

WORKED EXAMPLE **PAST PAPER**

November 2025 • Paper 2H • Q3 • 4 marks

A box has 9 red, 43 blue and some yellow bricks. $\frac{7}{20}$ are yellow. Each brick weighs 35 g. Work out the total weight of the yellow bricks.

STEP 1 $9 + 43 = 52$ is $\frac{13}{20}$ ◀ known fraction

STEP 2 $52 \div 13 = 4$ bricks per part ◀ one part

STEP 3 $7 \times 4 = 28$ yellow ◀ scale up

STEP 4 $28 \times 35 = 980$ ◀ weight

ANSWER 980 g

✓ total 80 bricks: $\frac{28}{80} = \frac{7}{20}$ ✓

WATCH OUT

✗ $\frac{7}{20} \times 52$

✓ 52 is $\frac{13}{20}$, not the whole

The 52 bricks are the non-yellow ones.

EXAM TIP

Check at the end: does your yellow count really make $\frac{7}{20}$ of the total?

Check yourself

calculator first • answers on page 17

6.1 Dr Eslam

Sam spends $\frac{1}{4}$ on food and $\frac{2}{5}$ on rent. He has \$70 left. How much did he start with?

Answer:

6.2 Practice • Hard

$\frac{7}{15}$ of some money is \$420. How much is the whole amount?

Answer:

6.3 Dr Eslam

$\frac{3}{5}$ of a class of 30 are girls; $\frac{1}{3}$ of the girls wear glasses. How many girls wear glasses?

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

1.1 $\frac{31}{8}$

1.2 14 kg

1.3 $\frac{3}{4}$

2.1 $\frac{70}{15} + \frac{57}{15} = \frac{127}{15} = 8\frac{7}{15}$

2.2 $\frac{32}{12} + \frac{45}{12} = \frac{77}{12} = 6\frac{5}{12}$

2.3 $\frac{28}{12} + \frac{21}{12} = \frac{49}{12} = 4\frac{1}{12}$

3.1 $\frac{112}{21} - \frac{60}{21} = \frac{52}{21} = 2\frac{10}{21}$

3.2 $1\frac{13}{20}$

3.3 $2\frac{11}{12}$

4.1 $\frac{16}{5} \times \frac{21}{8} = \frac{42}{5} = 8\frac{2}{5}$

4.2 $\frac{12}{7} \times \frac{35}{16} = \frac{15}{4} = 3\frac{3}{4}$

4.3 $\frac{15}{4} \times \frac{7}{9} = \frac{35}{12} = 2\frac{11}{12}$

5.1 $\frac{14}{3} \times \frac{5}{6} = \frac{35}{9} = 3\frac{8}{9}$

5.2 $\frac{7}{3} \times \frac{4}{21} = \frac{4}{9}$

5.3 $\frac{27}{4} \times \frac{7}{18} = \frac{21}{8} = 2\frac{5}{8}$

6.1 \$200

6.2 \$900

6.3 6

THE WHOLE TOPIC ON ONE PAGE

Fractions cheat sheet



Your notes
V1 Basics

$$\frac{3}{8} \text{ of } 240 = 90$$

÷ bottom, × top.

V2 Add 3/17

$$\frac{78}{21} + \frac{35}{21} = 5\frac{8}{21}$$

Improper → same bottom → add tops.

V3 Subtract 2/17

$$\frac{154}{21} - \frac{75}{21} = 3\frac{16}{21}$$

Improper — no borrowing.

V4 Multiply 6/17

$$\frac{18}{7} \times \frac{28}{9} = 8$$

Cancel, then top × top.

V5 Divide 5/17

$$\frac{24}{7} \times \frac{3}{8} = 1\frac{2}{7}$$

KFC: keep · flip · change.

V6 Problems 1/17

$$52 \div 13 = 4$$

Find one part, then scale.

**TOP 3 LOST MARKS**

1. adding the bottoms ×
2. multiplying wholes and parts separately ×
3. flipping the wrong fraction ×

BOTTOM LINE Mixed → improper → ±: same bottom · ×: cancel · ÷: KFC → back to mixed = the given answer.